

1 How reviews are discussed and decided

Question 10 in the open questions asks for this directly: “Then try to play out potential discourses across the three reviewers, and the chair as well. This aspect is more subtle. Some research on this topic, and even specific research about paper review, rebuttal, and acceptance or rejection deliberation. I believe there is such research.”

There is. Three studies carry it, and two of them were run by program chairs on their own conference.

1.1 The consistency experiments

The design is the same in both years. A fraction of submissions is reviewed twice by two independent committees that do not know about the duplication, and the two outcomes are compared.

2021. “During the assignment phase of the review process, we chose 10% of papers uniformly at random,” assigning each two area chairs and twice the usual number of reviewers, with the committees split across two copies and unaware of it. 882 papers were duplicated out of 8,765 still under review.

The outcome: 203 of 882 papers, 23.0 percent, had inconsistent outcomes between the two committees. Excluding borderline cases the chairs marked as bumpable, the figure fell to 16 percent. And the number that matters for an author: “50.6% of papers accepted by the first committee were rejected by the second.”

When the chairs asked what the disagreements were about, the largest single answer was that there was nothing to disagree about. “48% of multiple-choice responses were ‘This was a borderline paper that could have gone either way; there was no real disagreement between the committees.’” One area chair called it noise on the decision frontier.

2014, revisited seven years later. Cortes and Lawrence went back to the first experiment with the citation record in hand. “We determine that 50% of the variation in reviewer quality scores was subjective in origin.” For accepted papers there was no correlation between quality scores and later impact measured by citations. For rejected papers, traced to wherever they were eventually published, there was a correlation.

Their conclusion is the sentence to keep: “the reviewing process for the 2014 conference was good for identifying poor papers, but poor for identifying good papers.”

They also warn against the obvious fix: they “warn against removing the subjective element.”

1.2 What moves during the discussion

The consistency experiments measure whether the outcome is stable. They do not measure what changes it. A process-centric study does.

Jung, Pyeon, Heo, and Ahn examined ICLR from 2017 to 2025, “encompassing over 28,000 submissions,” using the record that open review platforms create: “not only scores and comments but also rebuttals, reviewer-author exchanges, reviewer disagreements, and meta-reviewer decisions.”

Their finding on the response phase: “the rebuttal stage emerges as a decisive phase: timely, substantive, and interactive author-reviewer communication strongly increases the likelihood of

acceptance, often outweighing initial reviewer skepticism.”

Alongside it, and in the same sentence class, they name “effective resolution of reviewer disagreement” as correlating with higher acceptance, together with clearer writing, balanced visual presentation, and earlier submission.

The dimension they measure for the committee side is “the patterns of reviewer disagreement and meta-review mediation (e.g., score variance, confidence weighting).”

1.3 The score movement figures already collected here

The tips work for this package recorded the movement data separately, and it fits the picture above. Most scores do not move: between 75 and 81 percent unchanged, 17 to 23 percent up, and about 1 percent down. But acceptance conditions sharply on movement: between 55.7 and 57.6 percent acceptance where a score moves, against 7.8 to 12.4 percent where none does.

Read with the consistency figures, that says the response is not competing against a fixed judgment. It is competing against a decision that was close to arbitrary in about half of frontier cases, where a single moved score changes the acceptance probability by a factor of five or more.

1.4 What follows for a response

Five consequences, and they are about who the reader is rather than about what to argue.

The reader who decides is the chair, not the reviewer. The chair mediates disagreement and weights by confidence. A response that answers three reviewers separately and never gives the chair a way to reconcile them has left the mediation undone.

A split panel is the leverage point, not the problem. Where two reviewers disagree about the same object, one of them has already made the argument. The wording checklist for this package states the move as a rule: use another reviewer’s assessment as evidence where they disagree with each other. It costs no words of the author’s own credibility.

At the frontier, the goal is to give a reason, not to win. Forty eight percent of disagreements were reported as having no real disagreement behind them. In that regime the response supplies the sentence a chair can quote when writing the decision.

Expertise is weighted, so it matters which reviewer holds which position. Confidence weighting is one of the measured dimensions. A concern raised by a self-declared expert and a concern raised by a reviewer with some familiarity do not carry the same weight, and a response that spends equally on both has misallocated.

The process detects bad work better than it detects good work. Cortes and Lawrence found the 2014 process good at identifying poor papers and poor at identifying good ones. The operational reading: removing a reason to reject is worth more than adding a reason to accept, because the process is more reliable in that direction.

1.5 The configuration in front of us

This package’s own panel is the frontier case the experiments describe. One accept from a self-declared expert, one weak accept from a knowledgeable reviewer, one weak reject from a reviewer with some familiarity.

Two of the comments are about the same object and point in opposite directions. One reviewer infers a simplistic environment from the small training budget. A different reviewer credits the same testbed for running VMs, real services, real network isolation, and correlated user activity, and for being explicit about the boundary between real and emulated components.

That pair is the disagreement the chair has to mediate, and the second reviewer has already written the answer to the first.

1.6 Sources

- The NeurIPS 2021 Consistency Experiment, posted on the conference blog by the program chairs, 8 December 2021. Read as a web page. All 2021 figures above are quoted from it.
- Corinna Cortes and Neil D. Lawrence, *Inconsistency in Conference Peer Review: Revisiting the 2014 NeurIPS Experiment*, arXiv 2109.09774. **Abstract only.** The two findings quoted are from the abstract. The disagreement rate for 2014 that circulates in secondary summaries is not in the abstract and is not repeated here.
- Sangkeun Jung, Goun Pyeon, Inbum Heo, and Hyungjin Ahn, *What Drives Paper Acceptance? A Process-Centric Analysis of Modern Peer Review*, arXiv 2509.25701. **Abstract only.** Every quotation is from the abstract, and the effect sizes behind the claims have not been checked.
- The score movement figures reach this file through the tips work already done for this package, which records the full citations.

The five consequences and the reading of this package’s own panel are extensions. None of the sources discusses this submission.